

1. **Explain the relationship between `A.txt` and its inverse. What kind of matrix is `A.txt`?**
 - a. The matrix in `A.txt` is a Pascal Matrix, which means that all elements above the diagonal are 0. The relationship is that the inverse contains coefficients with altering signs.
2. **Explain the relationship between `B.txt` and its inverse. What kind of matrix is `B.txt`?**
 - a. That matrix in `B.txt` is a spiral matrix (values are in a clockwise pattern). The relationship is that the inverse only exists if the determinant is non-zero.
3. **What was your approach to solve Part 3?**
 - a. Part 3 was super tricky to solve. I first noticed after several (which is an understatement) hours that there is likely a character issue in the given text file (unallowed characters; which was determined to be the comma). So, I used `Ada.Characters.Handling` to filter out non-graphic characters before parsing the line. (Essentially, used the comma as a delimiter instead of regex since I had trouble getting it to work)
 - b. I wrote a `Sanitize_Number` function to extract only digits and the minus sign from each token.
 - i. (This could have been done with regex, but it was easier for me to do it this way and use the comma as a delimiter)
 - c. After converting each value into Floats in 'X' and 'Y' Arrays, the points were now ready to be calculated.
 - d. According to the internet, the formula I applied is called the shoelace formula, defined in the Lab Instructions. I looped over every point and tracked the coordinates with the largest distance.

- e. After this, I tracked the coordinates and used them to display and calculate the largest area.
4. What is the correct value of `det(m3)` in Part 4?
- a. The correct value of `det(m3)` is the integer value (not the online calculator) as the matrix contains only whole numbers (integers). The decimal error is due to floating-point rounding errors as the calculator is likely using floating-point operations.
5. Explain in your own words why there is a difference between the odin calculation and the online calculator described in Part 4.
- a. The difference is that the online calculator likely uses floating-point arithmetic which allows for some small errors in calculations. The calculator on ODIN uses Integers which do not allow for the error that floating-point numbers do.